Ahmedabad University

CSE 641 Computer Vision

Weekly Report 4

Automated Segmentation of Post-Treatment Glioma Sub-regions and Resection Cavities from Multi-Parametric MRI.

Submitted to faculty: Prof. Mehul Raval

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Student Details

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Aim:

To conduct Exploratory Data Analysis (EDA) for multi-patients and train a baseline 3D U-Net for tumor sub-region segmentation.

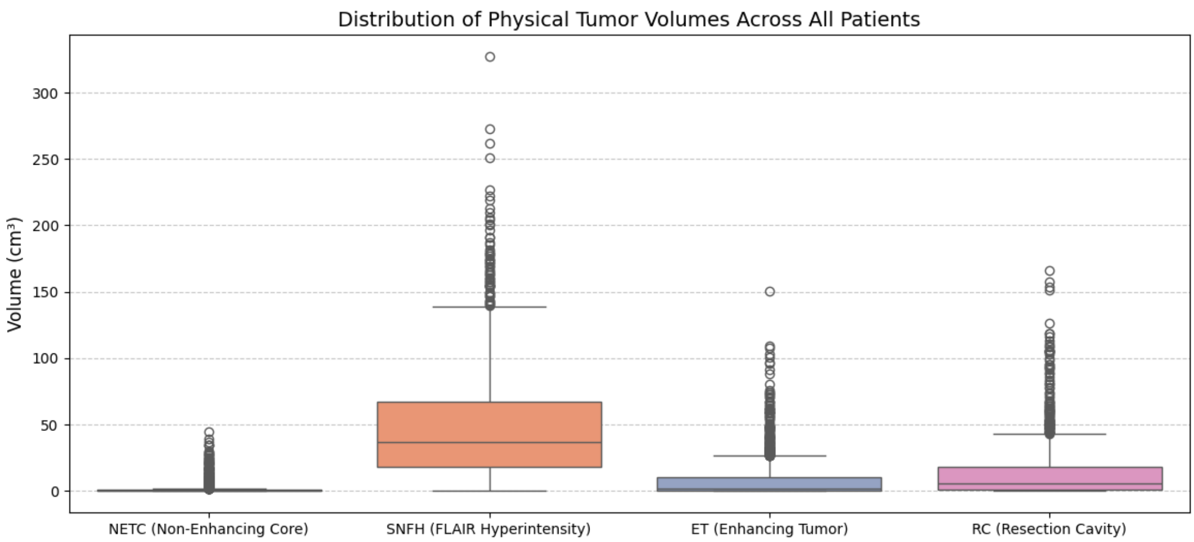
Introduction:

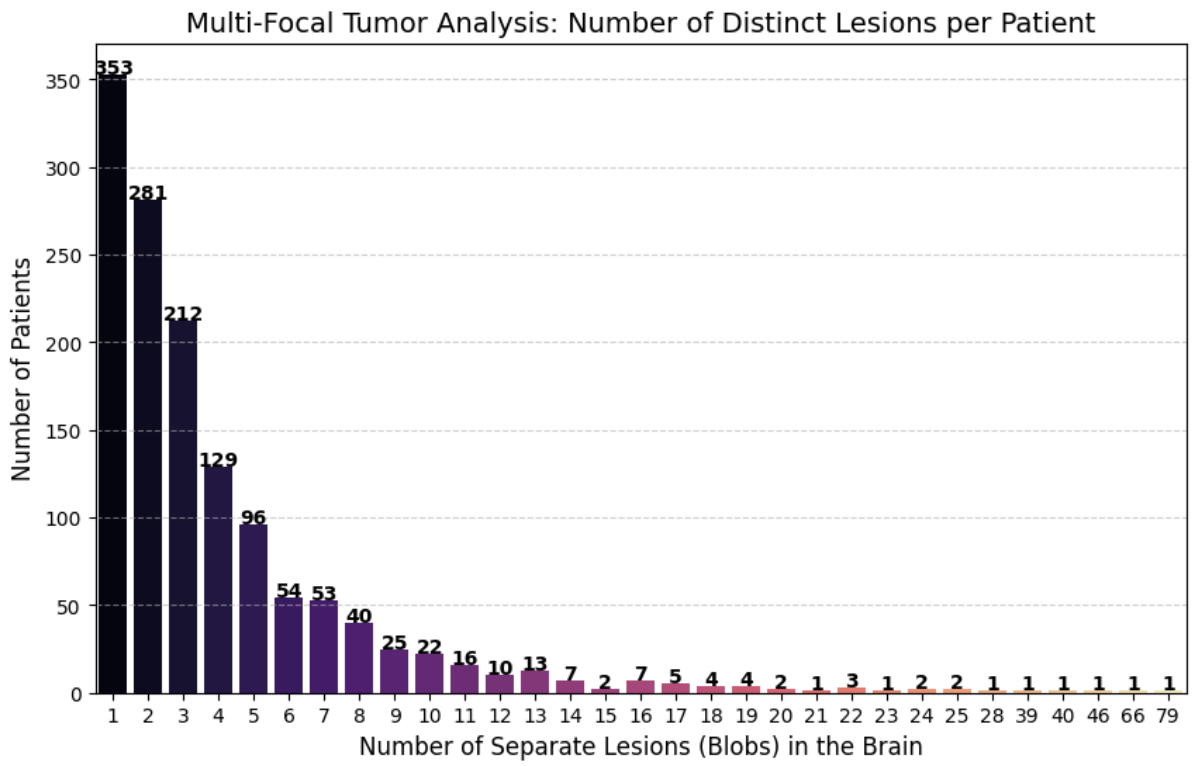
Diffuse gliomas are notorious and difficult to treat, and monitoring them post-surgery, chemotherapy, or radiation is complex due to the structural changes after treatments and varying tissue changes. During this week, we worked on building a strong base to deal with these morphological complexities. We are using the BraTS-GLI dataset (1,350 cases) and the MONAI library to create a system that can assist clinicians by giving them accurate volumetric reports of brain scans taken after surgery.

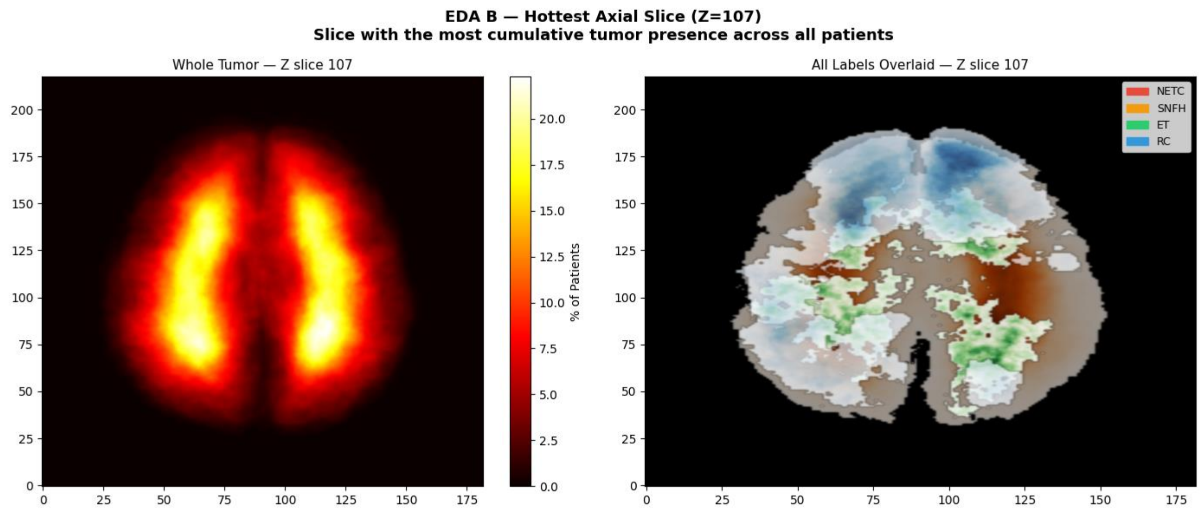
Work Completed:

- **Exploratory Data Analysis (EDA):**
    - Carried out volumetric analysis, where SNFH (Edema) is found to be the most dominant region, and NETC is found to be absent in 58.1% of cases.
    - Carried out 3D connected component analysis, where 7.3% of patients have multi-focal pathology.
    - Mapped tumor centroids and found a peak frequency at axial slice position Z = 107.

Results:







- **Preprocessing Pipeline:**
  - Implemented a custom 3D multimodal data loader using PyTorch and MONAI.
  - Implemented channel-wise non-zero intensity normalization and foreground cropping for optimal GPU memory.
  - Implemented a targeted 96 x 96 x 96 voxel patch extraction strategy.
- **Baseline Model Implementation:**
  - Implemented a 3D U-Net architecture using residual units and five levels of resolution (channels: 16-256).
  - Implemented Dice Loss using softmax activation and Adam optimizer (learning rate: $1 \times e^{-4}$).
  - Verified the pipeline using a validation set of 270 patients.

Next steps and goals:

Architecture Transition: Transition from the 3D U-Net base architecture to a sophisticated SegResNet architecture.

Refining Inference: Refine the sliding window inference such that the crop size during training is comparable to the window size during inference.

Conclusion:

This week, we were able to accomplish our goal of performing a comprehensive Exploratory Data Analysis (EDA) and creating a baseline 3D U-Net model. We learned important insights from the EDA, particularly on tumor morphology, where we observed that the SNFH region is dominant and that most of the data is multi-focal in nature. We were able to develop an effective preprocessing pipeline and validate our baseline model using MONAI and PyTorch, and this will help us in creating a robust segmentation system. We will be able to easily switch to a more sophisticated SegResNet model in the future for more accurate volumetric reporting.

References:

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5. A. Ferreira et al., “Improved Multi-Task Brain Tumour Segmentation with Synthetic Data Augmentation,” arXiv preprint arXiv:2411.04632, 2024.
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